

# UG10351

## RTD on the S32G A53 – Quick Start Guide

Rev. 1.0 — 12 November 2025

User guide

### Document information

| Information | Content                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keywords    | S32G, Bootloader, M7_0 application, M7_1 application, A53 RTD application, A53 BSP                                                                                                                                                                               |
| Abstract    | This documentation provides guidance for users implementing NXP RTD on the S32G A53. It outlines the complete process, from environment setup to application execution. The project demonstrates that NXP RTD drivers function effectively on the S32G A53 core. |



## 1 Introduction

This documentation provides guidance for users implementing NXP RTD on the S32G A53. It outlines the complete process, from environment setup to application execution. The project demonstrates that NXP RTD drivers function effectively on the S32G A53 core.

## 2 High-level architecture

This project is a multi-core demonstration that includes the bootloader, M7\_0 application, M7\_1 application, A53 RTD application, and A53 BSP.

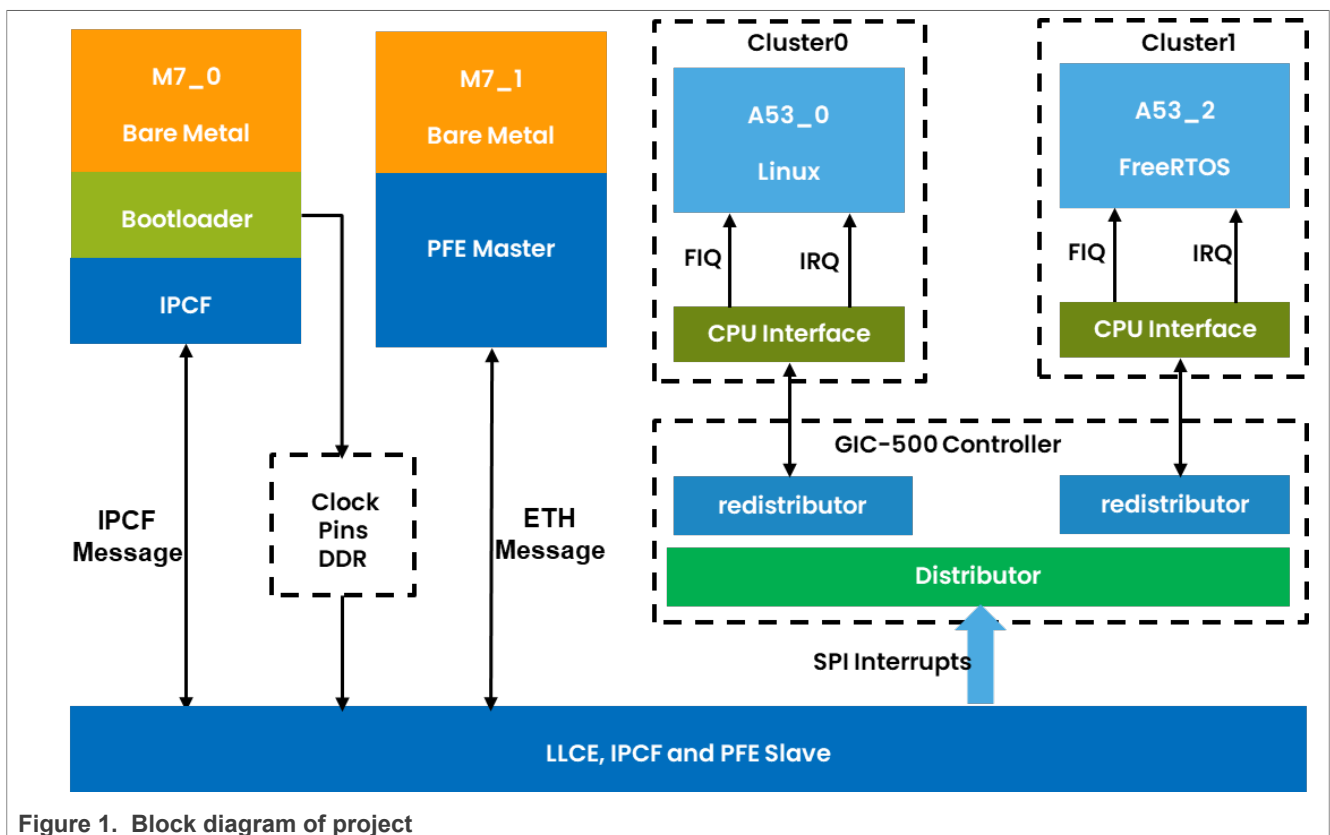


Figure 1. Block diagram of project

## 3 Basic boot process

According to the bootloader startup rules, the M7\_0 core is started last, while the other cores are started simultaneously. However, because Linux booting on the A53\_0 core requires passing through the ATF, U-Boot, and Kernel stages, A53\_0 must be started later than A53\_2. This sequence can lead to conflicts over several shared resources. Therefore, patches must be applied to the BSP image during compilation to prevent these conflicts.

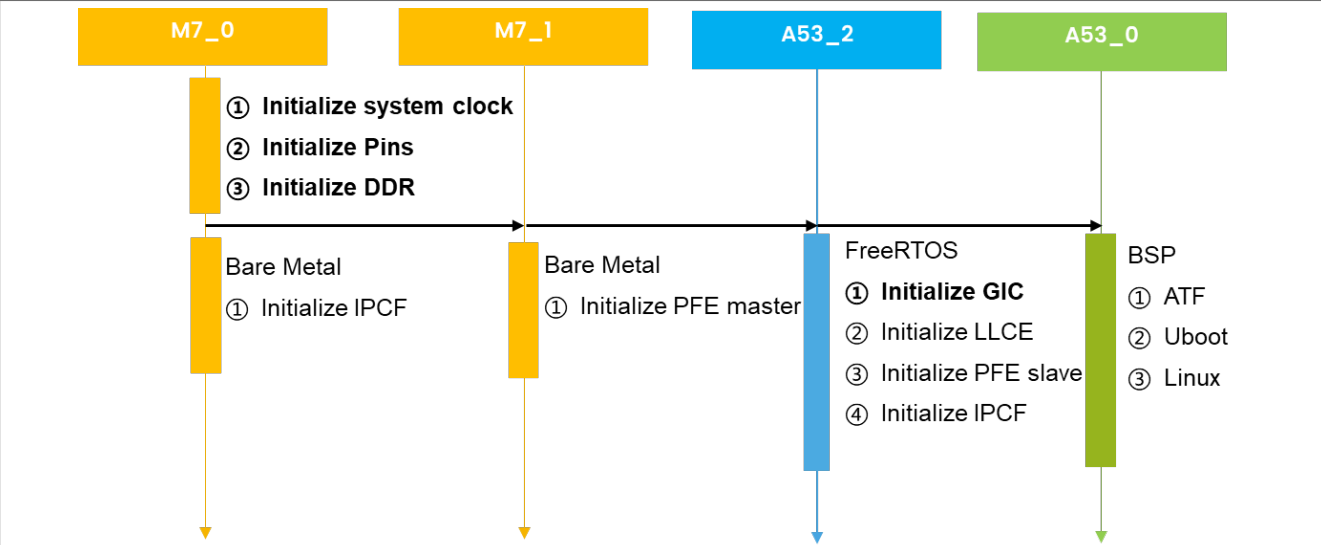


Figure 2. Booting process

1. M7\_0 core integrates a bootloader that is responsible for booting the M7\_1, A53\_0, and A53\_2 cores. The bootloader pre-initializes the clock, pins, and DDR for the A53\_2 core to prevent resource conflicts among different cores. The M7\_0 application runs an IPCF instance to receive messages from the A53 real-time application.
2. M7\_1 executes the PFE Master example.
3. A53\_2 runs LLCE, IPCF, and PFE Slave RTD drivers, and it initializes the GIC distributor to route interrupts to different A53 cores. It also resolves unaligned memory access issues when RTD runs on AArch64 systems and integrates the FreeRTOS real-time operating system.
4. A53\_0 runs the BSP, handling conflicts in the GIC distributor between ATF and the kernel, as well as resolving clock, pin, and CANPHY conflicts between the bootloader and Linux.

## 4 Hardware and software prerequisites

This chapter describes the hardware and software prerequisites necessary to use the demo.

### 4.1 Hardware prerequisites

Table 1. Hardware prerequisites

| Name                       | Location                                                                                                                                        |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| S32G RDB2/RDB3             | <a href="https://www.nxp.com/S32G-VNP-RDB2">https://www.nxp.com/S32G-VNP-RDB2</a>                                                               |
| TRACE32 Power Debug System | <a href="https://www.lauterbach.com/supported-platforms/architectures/arm">https://www.lauterbach.com/supported-platforms/architectures/arm</a> |

### 4.2 Software prerequisites

Table 2. Software prerequisites

| Name               | Version | Location                                                                          | Description                                                                 |
|--------------------|---------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| S32G2/G3 Linux BSP | 43.0    | <a href="https://github.com/nxp-auto-linux">https://github.com/nxp-auto-linux</a> | Linux Board Support Package contains ARM Trusted Firmware and Linux Kernel. |

Table 2. Software prerequisites...continued

| Name                                        | Version   | Location                                                                                                                                                                                                                                                                                                                                  | Description                                                                                                                                                                                                                       |
|---------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GoldVIP for S32G2/G3                        | 1.13.0    | Log in to nxp.com and go to:<br>> Software Licensing and Support<br>> Software accounts<br>> Automotive SW - S32G - Vehicle Integration Platform (GoldVIP)<br>> Automotive SW - S32G - Vehicle Integration Platform (GoldVIP)<br>> GoldVIP for S32G2/G3 1.13.0<br>> GoldVIP-S32G2/G3-1.13.0.zip                                           | S32G Vehicle Integration Platform for S32G contains various example for the S32G platform, along with the System- level Bootloader example.                                                                                       |
| S32G Real Time Drivers                      | 5.0.0     | > Software Licensing and Support<br>> Software accounts<br>> Automotive SW – S32G Standard Software<br>> Automotive SW - S32G - Real Time Drivers (RTD) (Cortex-M7)<br>> S32 Real-Time Drivers Version 5.0.0<br>> SW32G_RTD_4.4_5.0.0_D2410.exe                                                                                           | Real Time Drivers (RTD) is a set of AUTOSAR compliant device drivers responsible for hardware abstraction and enablement for the NXP targets.                                                                                     |
| S32G Inter Platform Communication Framework | 4.10.0    | Log in to nxp.com and go to:<br>> Software Licensing and Support<br>> Software accounts<br>> Automotive SW – S32G Standard Software<br>> Automotive SW - S32G - Inter-Platform Communication Framework (IPCF) (Cortex-A53 and Cortex-M7)<br>> S32G Inter Platform Communication Framework Version 4.10.0<br>> SW32G_IPCF_4.10.0_D2405.exe | Inter-Platform Communication Framework (IPCF) is a subsystem which enables applications, running on multiple homogenous or heterogeneous processing cores to communicate over various transport interfaces (i.e., shared memory). |
| LLCE Firmware and Driver                    | 1.0.10    | Log in to nxp.com and go to:<br>> Software Licensing and Support<br>> Software accounts<br>> Automotive SW – S32G Standard Software<br>> Automotive SW - S32G - LLCE Driver + Firmware<br>> S32G_LLCE_1.0.10<br>> S32G_LLCE_GATEWAY_1.0.10_D2505.exe                                                                                      | Low Latency Communication Engine (LLCE) firmware is intended to offload host applications regarding CAN routing and other time-consuming communication operations.                                                                |
| Packet Forwarding Engine driver (Cortex-M7) | 1.5.0     | Log in to nxp.com and go to:<br>> Software Licensing and Support<br>> Software accounts<br>> Automotive SW - S32G Standard Software<br>> Automotive SW - PFE Driver + Standard Firmware<br>> S32G PFE MCAL 4.4 driver - SW32G_MCL01_1.5.0_D2411<br>> PFE-DRV_S32G_M7_MCAL_1.5.0.zip                                                       | Packet Forwarding Engine (PFE) Driver for ARM Cortex-M7 processors.                                                                                                                                                               |
| Lwip                                        | 2.1.3     | <a href="https://savannah.nongnu.org/projects/lwip/">https://savannah.nongnu.org/projects/lwip/</a>                                                                                                                                                                                                                                       | Lwip is a small independent implementation of the TCP/IP protocol suite                                                                                                                                                           |
| TRACE32                                     | R.2024.02 | <a href="https://repo.lauterbach.com/download_trace32.html">https://repo.lauterbach.com/download_trace32.html</a>                                                                                                                                                                                                                         | Embedded debuggers and trace. The TRACE32 GUI can also be used as debug                                                                                                                                                           |

Table 2. Software prerequisites...continued

| Name                                                                    | Version     | Location                                                                                                                                                                                                                                              | Description                                                                                |
|-------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|                                                                         |             |                                                                                                                                                                                                                                                       | front-end for virtual targets, core simulators, and target GDB servers.                    |
| EB tresos Studio                                                        | 27.1.0      | Log in to nxp.com and go to:<br>> Software Licensing and Support<br>> Software accounts<br>> Automotive SW - S32G Reference Software<br>> Automotive SW - EB tresos Studio / AUTOSAR Configuration Tool<br>> EB tresos Studio 27.1.0                  | EB tresos Studio lets you configure, validate, and generate your ECU basic software (BSW). |
| S32 Design Studio 3.5 Development Package with support for S32G devices | 3.5.3_D2306 | Log in to nxp.com and go to:<br>> Software Licensing and Support<br>> Software accounts<br>> Automotive SW - S32G - S32 Design Studio<br>> S32 Design Studio 3.5.3 Development Package with support for S32G devices<br>> SW32G_S32DS_3.5.3_D2306.zip | S32 Design Studio 3.5 Development Package with support for S32G devices.                   |

## 5 Building project

### 5.1 How to rebuild Linux BSP image

The binary folder in the released software package already contains the pre-compiled BSP image. If recompilation is required, please refer to the compilation process described below.

#### 5.1.1 Adding ATF and kernel patch

1. Locate the arm-trusted-firmware directory in the YOCTO compiled directory. Then add 0001-atf-ddr-init.patch and 0001-atf-distributor-init.patch to the below directory.  
**/path/to/your/yocto/sources/meta-alb/meta-alb-bsp/recipes-bsp/arm-trusted-firmware/arm-trusted-firmware**
2. Add the patch files to the compilation process in the arm-trusted-firmware\_%.bbappend file.

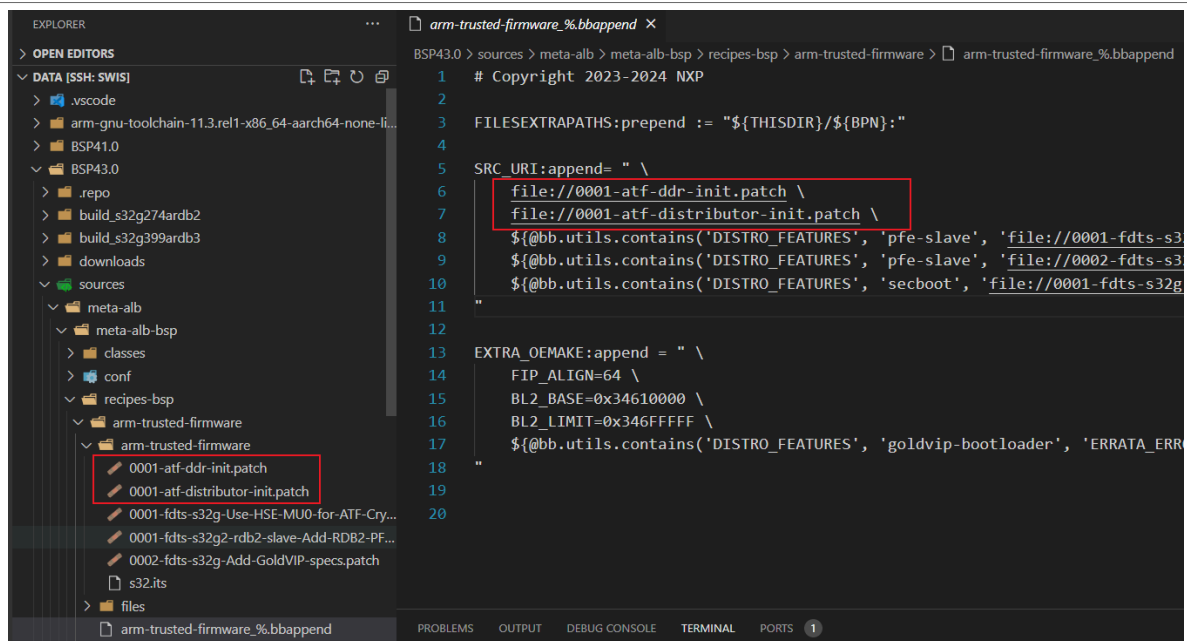


Figure 3. ATF patches add path

3. Locate the linux-s32 directory in the YOEOTO compiled directory. Then add 0001-bsp-distributor-init.patch and 0001-deferred\_probe\_work\_func\_exclude\_can\_phy.patch to the below directory.  
**/path/to/your/yocto/sources/meta-alb/meta-alb-bsp/recipes-kernel/linux/linux-s32**
4. Add the patch files to the compilation process in the linux-s32\_6.6.bb file.

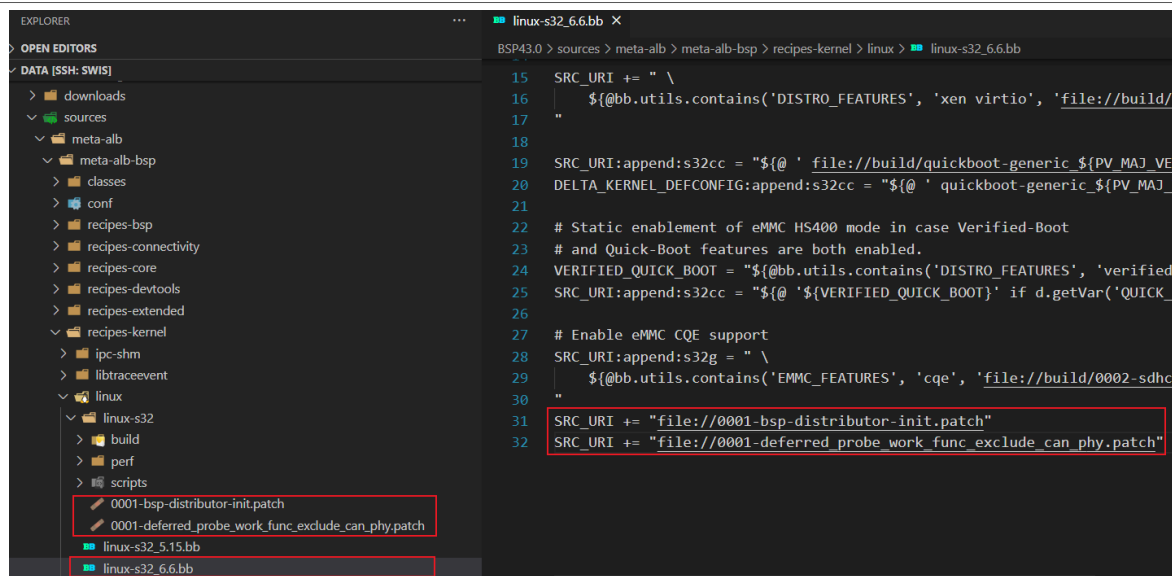


Figure 4. Kernel patches add path

### 5.1.2 U-boot and kernel menuconfig

1. Refer to the S32G2/G3\_LinuxBSP\_43.0\_User\_Manual for the u-boot menuconfig process and configure u-boot according to the configuration path provided below.

Table 3. Configuration path for u-boot

|    | Configuration path               |
|----|----------------------------------|
| -> | Device drivers                   |
| -> | Network device support           |
| -> | NXP S32G PFE Ethernet controller |
| -> | excludes                         |

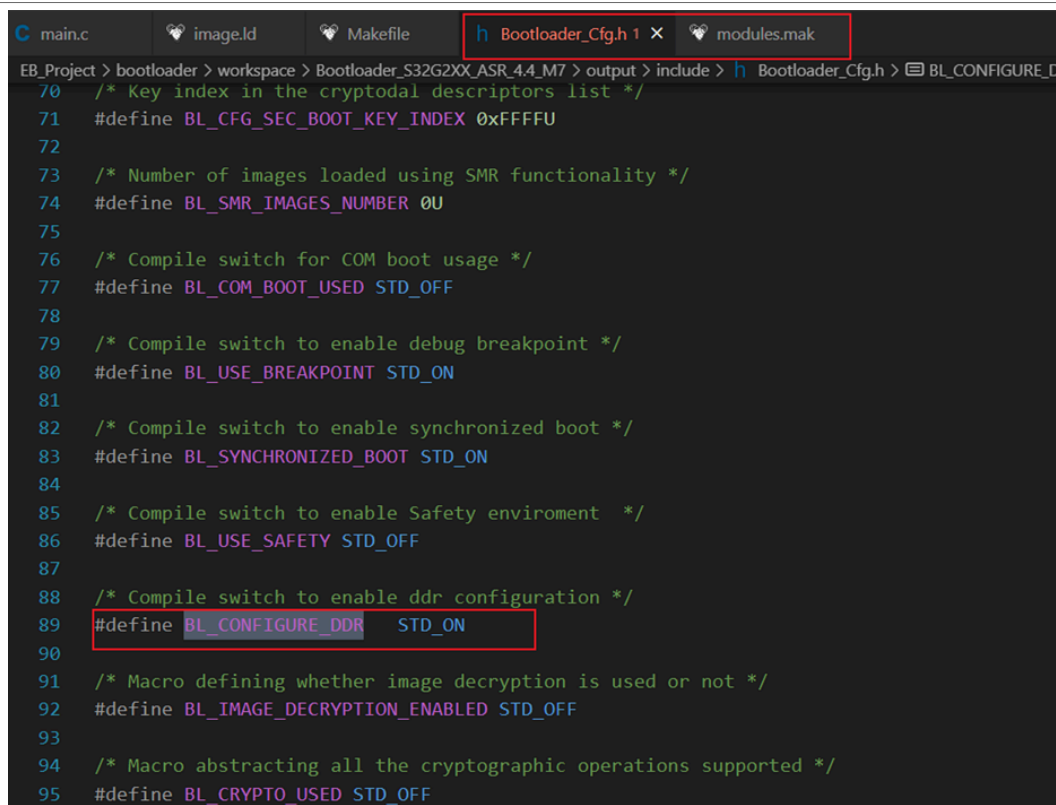
2. Refer to the S32G2/G3\_LinuxBSP\_43.0\_User\_Manual for the kernel menuconfig process and configure the kernel according to the configuration path provided below.

Table 4. Configuration path for kernel

|    | Configuration path |    | Configuration path                        |
|----|--------------------|----|-------------------------------------------|
| -> | Kernel Features    | -> | Device Drivers                            |
| -> | Max number of CPUs | -> | Network device support                    |
| -> | 2                  | -> | CAN Device Drivers                        |
| -> |                    | -> | Support for Freescale FLEXCAN based chips |
| -> |                    | -> | excludes                                  |

## 5.2 How to rebuild bootloader

1. Configure the paths of TRESOS\_DIR, MAKE\_DIR, GCC\_DIR, SRC\_PATH\_DRIVERS, DDR\_TOOL\_CONFIG\_DIR, and TRESOS\_WORKSPACE\_DIR in the bootloader compilation configuration file.
2. Every time you generate the configuration using EB, it is necessary to manually set BL\_CONFIGURE\_DDR from STD\_OFF to STD\_ON.



```
70 /* Key index in the cryptotal descriptors list */
71 #define BL_CFG_SEC_BOOT_KEY_INDEX 0xFFFFU
72
73 /* Number of images loaded using SMR functionality */
74 #define BL_SMR_IMAGES_NUMBER 0U
75
76 /* Compile switch for COM boot usage */
77 #define BL_COM_BOOT_USED STD_OFF
78
79 /* Compile switch to enable debug breakpoint */
80 #define BL_USE_BREAKPOINT STD_ON
81
82 /* Compile switch to enable synchronized boot */
83 #define BL_SYNCHRONIZED_BOOT STD_ON
84
85 /* Compile switch to enable Safety enviroment */
86 #define BL_USE_SAFETY STD_OFF
87
88 /* Compile switch to enable ddr configuration */
89 #define BL_CONFIGURE_DDR STD_ON
90
91 /* Macro defining whether image decryption is used or not */
92 #define BL_IMAGE_DECRYPTION_ENABLED STD_OFF
93
94 /* Macro abstracting all the cryptographic operations supported */
95 #define BL_CRYPTO_USED STD_OFF
```

Figure 5. Bootloader generates file configuration modification

3. Open S32DS 3.5 and create a new blank project, then use the IVT tool in S32DS to configure it as shown in the figure below. The QSPI reconfiguration parameters and DCD binary can be found in the **bootloader\framework\swc\bootloader\platforms\S32G3XX\res\flash** directory. If it is G2, set both the RAM start pointer and RAM entry pointer to 0x34700000. If it is G3, set both the RAM start pointer and RAM entry pointer to 0x34530000.



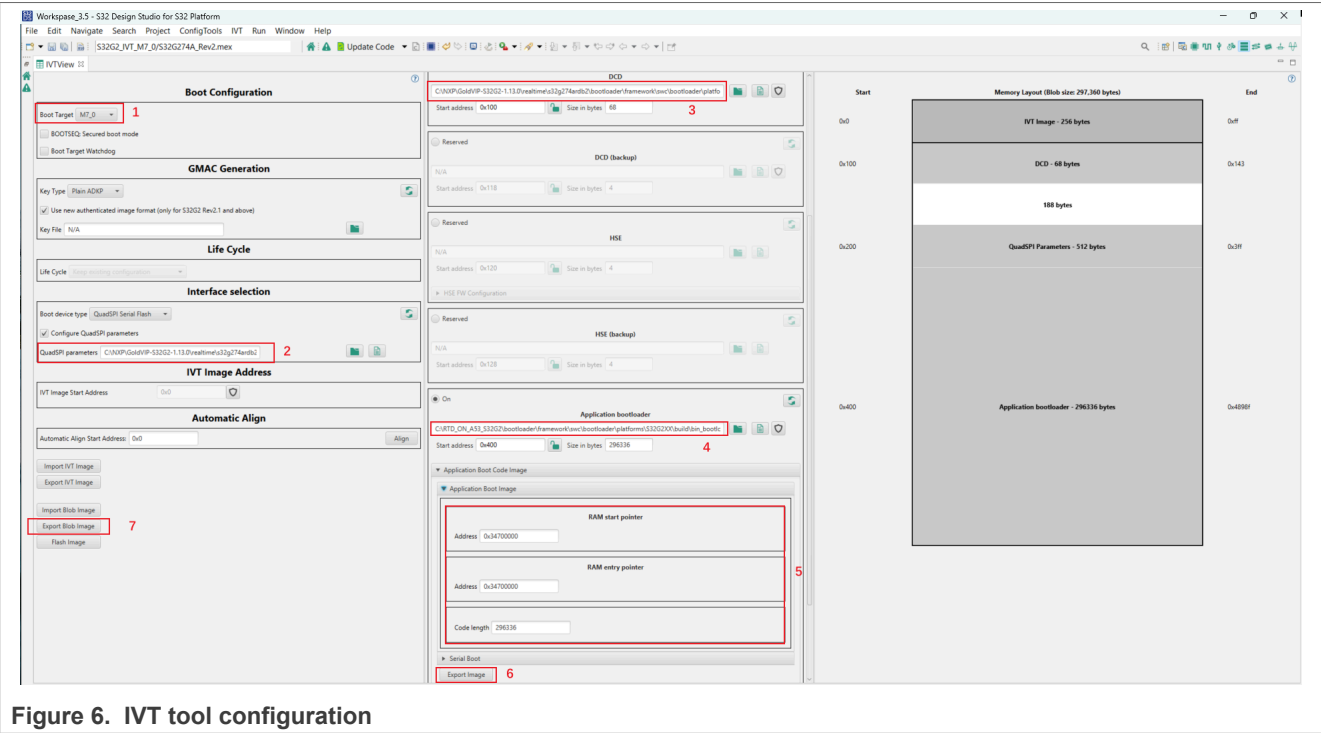


Figure 6. iVT tool configuration

### 5.3 How to rebuild IPCF

1. Download the SW32G\_IPCF\_4.10.0\_D2405 software package and modify the source files in the plugins as below.

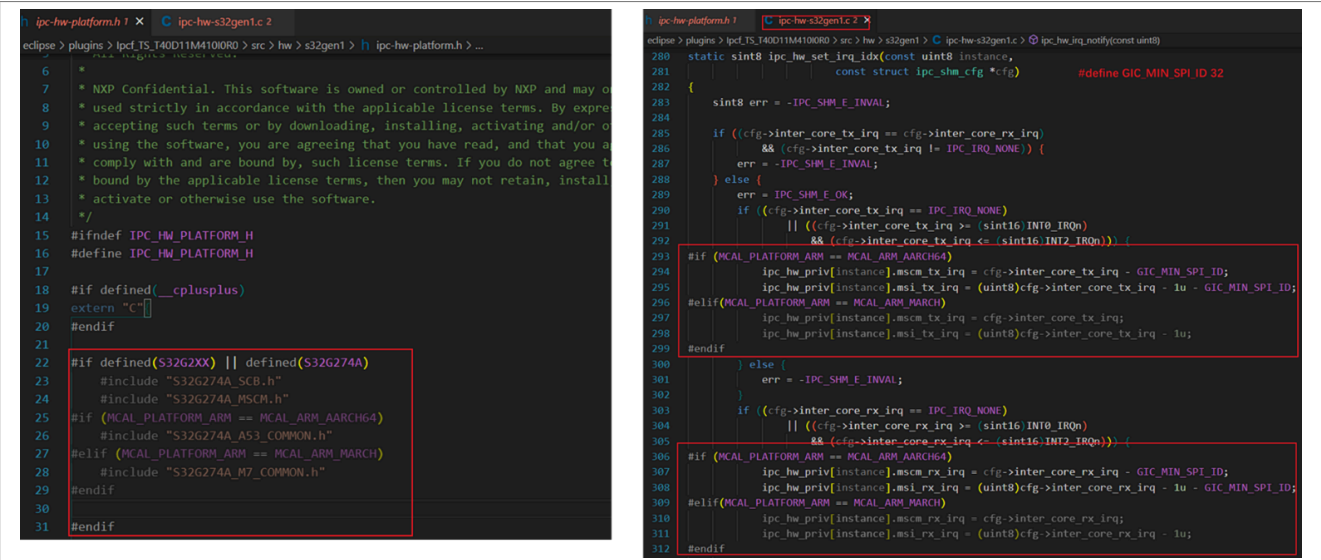
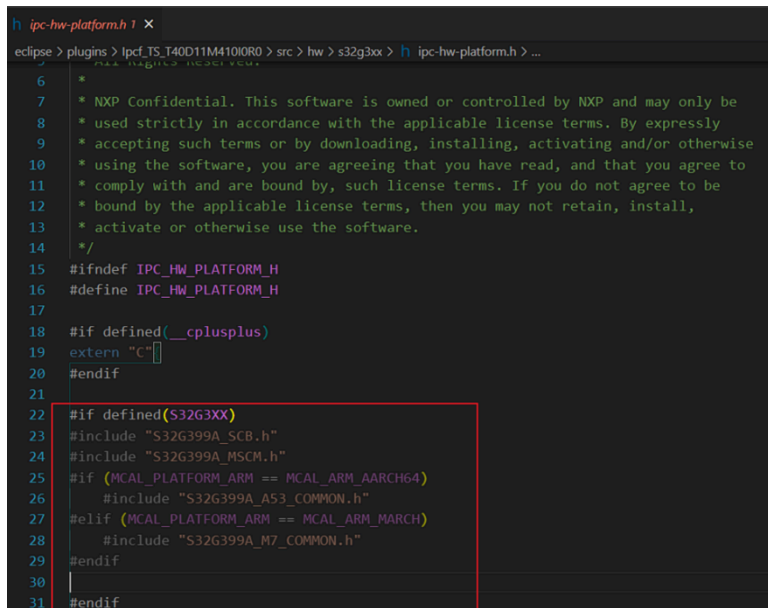


Figure 7. S32G2 IPCF source file modification



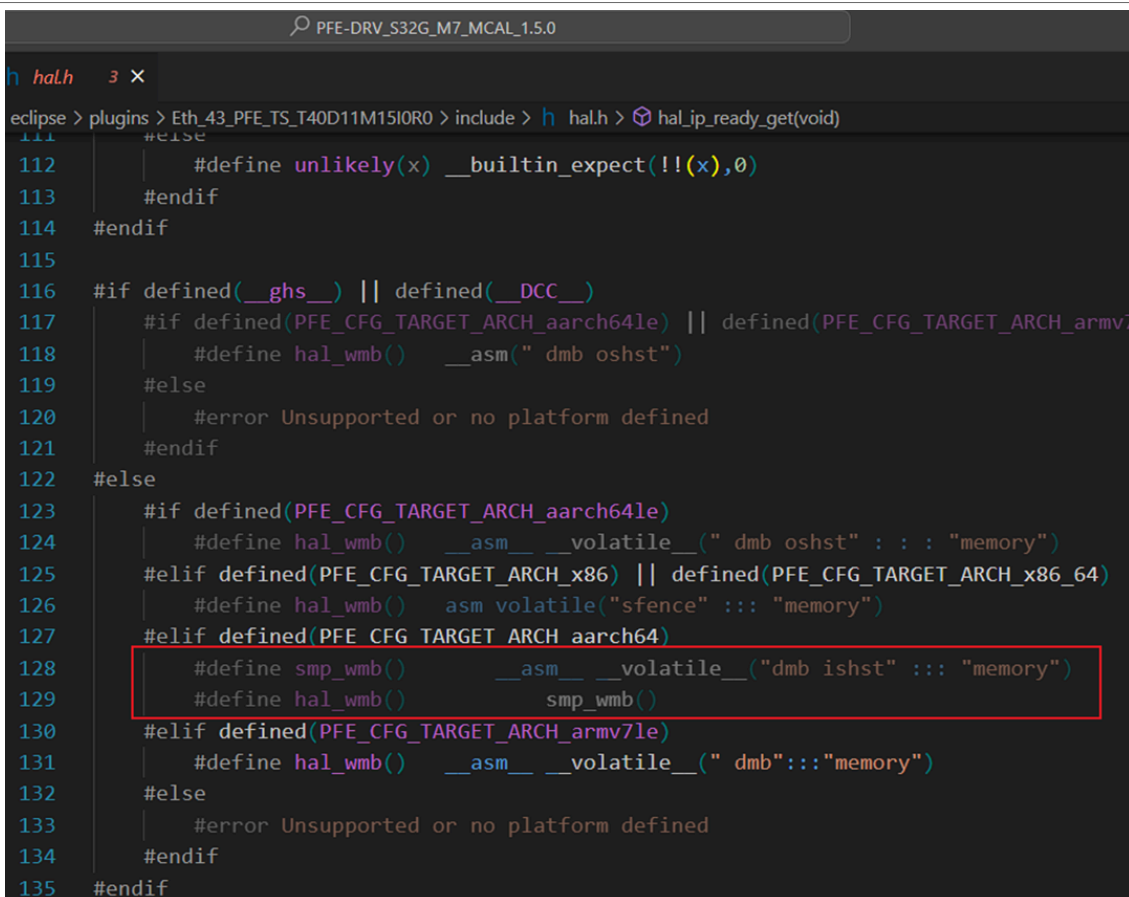
```
h ipc-hw-platform.h 1 x
eclipse > plugins > lpcf_TS_T40D11M410I0R0 > src > hw > s32g3xx > h ipc-hw-platform.h > ...
6  *
7  * NXP Confidential. This software is owned or controlled by NXP and may only be
8  * used strictly in accordance with the applicable license terms. By expressly
9  * accepting such terms or by downloading, installing, activating and/or otherwise
10 * using the software, you are agreeing that you have read, and that you agree to
11 * comply with and are bound by, such license terms. If you do not agree to be
12 * bound by the applicable license terms, then you may not retain, install,
13 * activate or otherwise use the software.
14 */
15 #ifndef IPC_HW_PLATFORM_H
16 #define IPC_HW_PLATFORM_H
17
18 #if defined(__cplusplus)
19 extern "C"
20 #endif
21
22 #if defined(S32G3XX)
23 #include "S32G399A_SCB.h"
24 #include "S32G399A_MSCM.h"
25 #if (MCAL_PLATFORM_ARM == MCAL_ARM_AARCH64)
26 #include "S32G399A_A53_COMMON.h"
27 #elif (MCAL_PLATFORM_ARM == MCAL_ARM_MARCH)
28 #include "S32G399A_M7_COMMON.h"
29 #endif
30
31 #endif
```

Figure 8. S32G3 IPCF source file modification

2. Open the project\_parameters.mk file in **your\path\M7\_0\_Source\project\_parameters.mk**. Set the GCC\_DIR, TRESOS\_DIR, PLUGINS\_DIR, and SHM\_DRIVER\_PATH variables to match your local directory structure. Once the paths are correctly configured, proceed to build the IPCF project.

## 5.4 How to rebuild PFE Master

1. Download the PFE-DRV\_S32G\_M7\_MCAL\_1.5.0 software package and modify the source files in the plugins as below.



```

111 | #else
112 |     #define unlikely(x) __builtin_expect(!!(x),0)
113 | #endif
114 #endif
115
116 #if defined(__ghs__) || defined(__DCC__)
117     #if defined(PFE_CFG_TARGET_ARCH_aarch64le) || defined(PFE_CFG_TARGET_ARCH_armv7
118         #define hal_wmb() __asm(" dmb oshst")
119     #else
120         #error Unsupported or no platform defined
121     #endif
122 #else
123     #if defined(PFE_CFG_TARGET_ARCH_aarch64le)
124         #define hal_wmb() __asm__ __volatile__ (" dmb oshst" ::: "memory")
125     #elif defined(PFE_CFG_TARGET_ARCH_x86) || defined(PFE_CFG_TARGET_ARCH_x86_64)
126         #define hal_wmb() asm volatile("sfence" ::: "memory")
127     #elif defined(PFE_CFG_TARGET_ARCH_aarch64)
128         #define smp_wmb() __asm__ __volatile__ ("dmb ishst" ::: "memory")
129         #define hal_wmb() smp_wmb()
130     #elif defined(PFE_CFG_TARGET_ARCH_armv7le)
131         #define hal_wmb() __asm__ __volatile__ (" dmb" ::: "memory")
132     #else
133         #error Unsupported or no platform defined
134     #endif
135 #endif

```

Figure 9. PFE MCAL driver modification

- Copy the Eth\_43\_PFE\_TS\_T40D11M15I0R0 folder to the plugins folder of RTD 5.0.0.
- Open the project\_parameters.mk file in **your\path\M7\_1\_Source\project\_parameters.mk**. Set the GCC\_DIR, TRESOS\_DIR, and PLUGINS\_DIR variables to match your local directory structure. Once the paths are correctly configured, proceed to build the PFE Master project.

## 5.5 How to rebuild A53 RTD project

### 5.5.1 LLCE rebuild process

Because of a bug in RTD 5.0.0, compiling LLCE 1.0.10 with RTD 5.0.0 results in LLCE Shared\_Memory being mapped to the wrong address. To resolve this, replace Can\_43\_LLCE\_MemMap.h in RTD 5.0.0 with the version from RTD 5.0.0 QLP03.

### 5.5.2 PFE rebuild process

- Each time you use EB to generate the PFE SLAVE configuration, the two generated files listed below need to be modified according to the picture shown below.

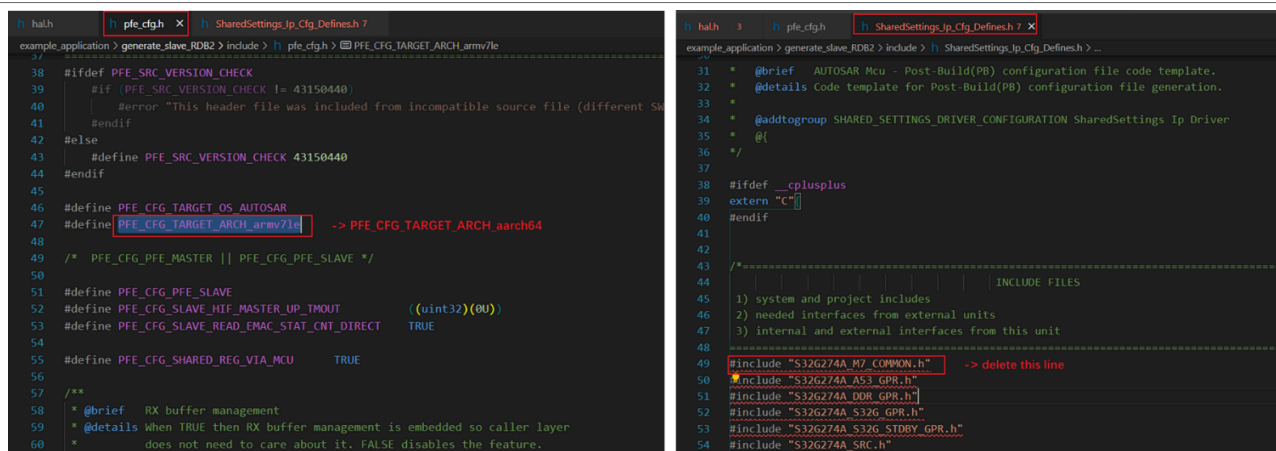


Figure 10. A53 RTD project generates file configuration modification

2. Open the config.mk file located at **your\path\RTOS\_AND\_RTD\_ON\_A53\config.mk**. Configure the paths for CrossToolDir, PLUGIN\_DIR, LLCE\_PLUGIN\_DIR, IPCF\_PLUGIN\_DIR, and LLCE\_BIN\_DIR according to your local environment. After updating these paths, build the A53 RTD project.

### 5.5.3 LWIP rebuild process

In the PFE SLAVE project, LWIP and PING\_TEST cannot run simultaneously. By default, the PFE SLAVE project enables the PING\_TEST function. If you do not need to use the LWIP function, please ignore this section.

1. Download the lwip 2.1.3 package from the official lwip website, then place the package in directory **your\path\RTOS\_AND\_RTD\_ON\_A53\PFE\_Integration**.
2. Change the compilation options in **your\path\RTOS\_AND\_RTD\_ON\_A53\config.mk** as below:

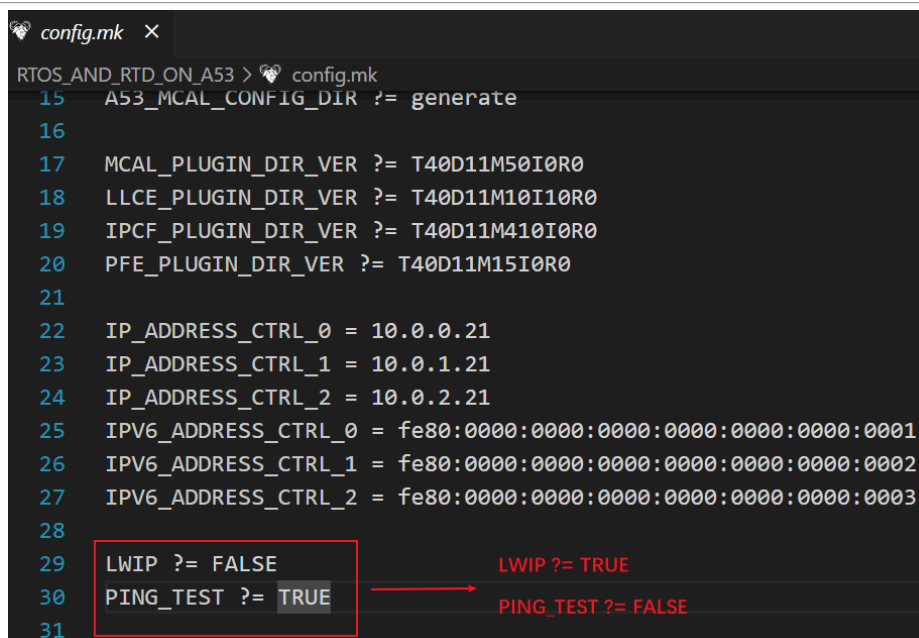


Figure 11. Lwip enable configuration

3. Recompile the project to generate a new RTD\_A53\_IMAGE.bin. Note that the size of the newly generated binary file will change, so you need to update the image size of A53\_2 in the Bootloader configuration accordingly and generate a new Bootloader\_Configuration.bin file.

## 6 Quick Start

### 6.1 Flash image via S32 Flash Tool

Refer to the QSPI memory layout below. Burn the images from the binary into the Flash in sequence, and then set the RCON of the RDB2/RDB3 board to NOR Flash boot mode.

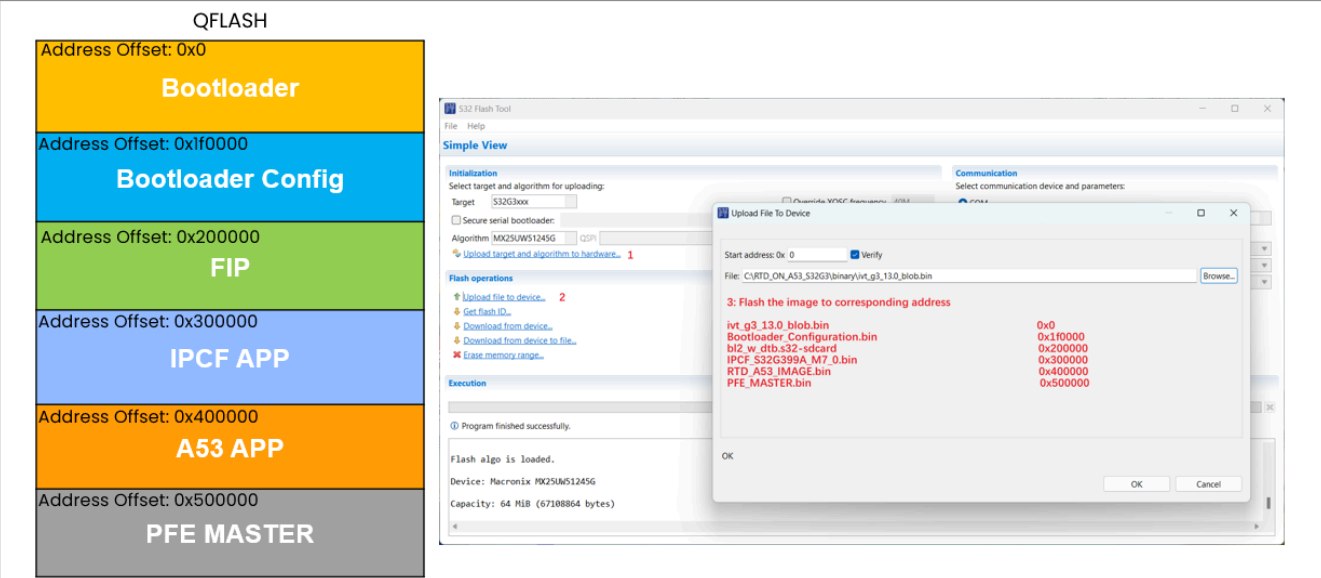


Figure 12. QSPI memory layout and burning steps

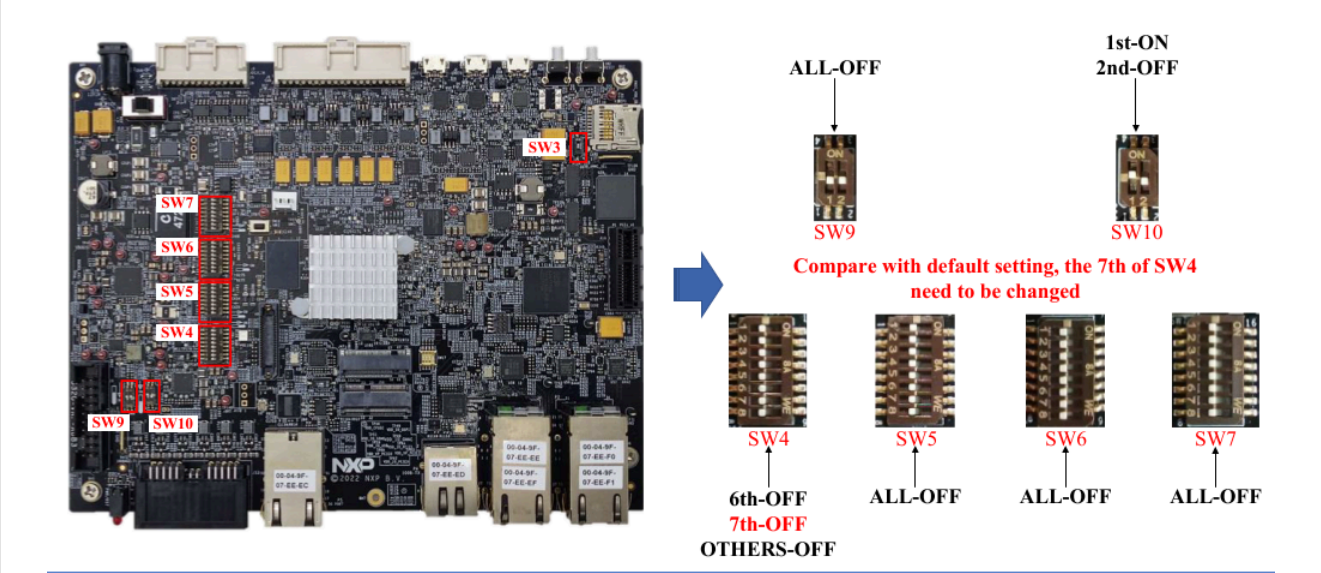


Figure 13. NOR Flash boot switch setting

### 6.2 SD card image burning

Prepare an SD card with sufficient memory, then use a card reader to mount it to the PC's file system. Next, use the following command to burn the BSP image. Make sure to insert the SD card into the target device after the burning process is complete.

```
dd if=/dev/zero of=/dev/sdb bs=512 count=1 && sync
```

```
dd if=fsl-image-auto-s32g399ardb3-xxx.rootfs.sdcard of=/dev/sdb bs=1M skip=4 seek=4 && sync
```

```
dd if=fsl-image-auto-s32g399ardb3-xxx.rootfs.sdcard of=/dev/sdb bs=1M count=4 && sync
```

## 7 Application

All tasks in this section have resolved conflicts between memory and peripheral interrupts, so they can run simultaneously without interfering with each other.

### 7.1 BSP running on A53\_0

#### 7.1.1 HW resources configured and used

UART0.

#### 7.1.2 Prerequisites

Connect UART0 to a host PC that has serial port software installed (e.g., MobaXterm).

#### 7.1.3 Functionality

After burning the BSP image with the added patch into the SD card, observe the serial port output. Once the system has started successfully, log in to the BSP through the console.

```
[ OK ] Started Network Name Resolution.
[ OK ] Reached target Network.
[ OK ] Reached target Host and Network Name Lookups.
[ OK ] Started Netperf Benchmark Server.
[ OK ] Started NFS status monitor for NFSv2/3 locking..
[ OK ] Started Respond to IPv6 Node Information Queries.
[ OK ] Started Network Router Discovery Daemon.
Starting Permit User Sessions...
[ OK ] Started Xinetd A Powerful Replacement For Inetd.
[ OK ] Finished Permit User Sessions.
[ OK ] Started Getty on tty1.
[ OK ] Started Serial Getty on ttyLF0.
[ OK ] Reached target Login Prompts.
[ OK ] Finished Load/Save Random Seed.
[ OK ] Started D-Bus System Message Bus.
[ OK ] Started User Login Management.
[ OK ] Reached target Multi-User System.
Starting Record Runlevel Change in UTMP...
[ OK ] Finished Record Runlevel Change in UTMP.

Auto Linux BSP 43.0 s32g399ardb3 ttyLF0

s32g399ardb3 login: root
root@s32g399ardb3:~# ls
root@s32g399ardb3:~# cd ..
root@s32g399ardb3:/home# cd ..
root@s32g399ardb3:/# ls
bin boot dev etc home lib lost+found media mnt opt proc run sbin srv sys tmp usr var
root@s32g399ardb3:/# █
```

Figure 14. BSP console log



## 7.2 LLCE Task

### 7.2.1 HW resources configured and used

LLCE CAN1 ~ LLCE CAN4

### 7.2.2 Prerequisites

Use the LLCE connectors listed in the [appendix](#) to externally connect LLCE CAN1 to CAN2, and LLCE CAN3 to CAN4.

### 7.2.3 Functionality

The LLCE task implements the CAN2CAN function. CANFD frames sent through LLCE CAN2 are received by LLCE CAN1 and internally routed to LLCE CAN4. LLCE CAN3 receives the routing frames from LLCE CAN4 and passes them to the A53 host. Attach the A53\_2 core, then enter the following command in the TRACE 32 command line:

**Data.Load.Elif ./Output/RTD\_ON\_A53.elf /GLOBTYPES /NoCode**

**Var.Watch Can\_RxIndication Can\_TxConfirmation**

Observe Can\_RxIndication and Can\_TxConfirmation changes in results.

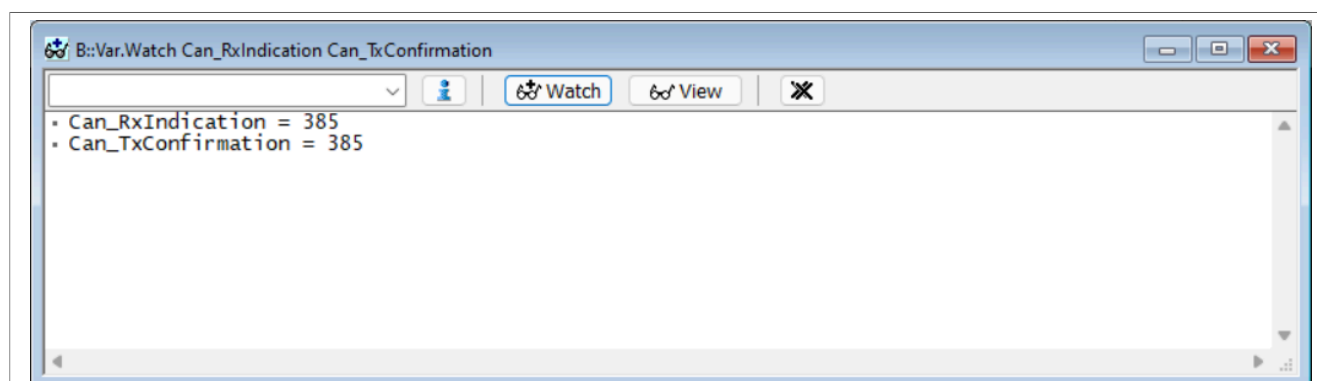


Figure 15. LLCE task Tx and Rx counter

## 7.3 IPCF Task

### 7.3.1 Prerequisites

The Cortex-M7\_0 core needs to run an IPCF instance to communicate with the Cortex-A53\_2 core.

### 7.3.2 Functionality

This task sends a message to the M7\_0 core via an interrupt to notify the remote core. After receiving the message, the M7\_0 core also sends a message back to the A53\_2 core via an interrupt. Attach the A53\_2 core, then enter the following command in the TRACE 32 command line:

**Data.Load.Elif ./Output/RTD\_ON\_A53.elf /GLOBTYPES /NoCode**

**Var.Watch app.tx\_count app.rx\_count**

You can observe the IPCF tx and rx counters by checking the app parameters. Additionally, you can dump the address 0x81500000 to view the reply message content received by the shared memory from M7\_0.

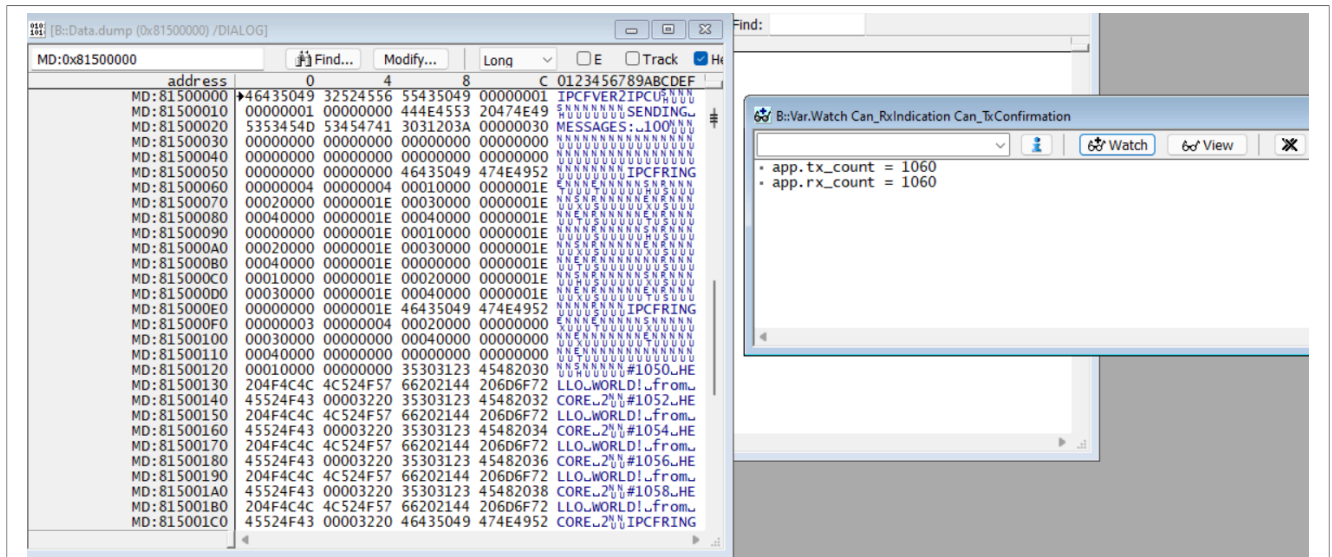


Figure 16. IPCF Task Tx and Rx counter

## 7.4 PFE Task

### 7.4.1 HW resources configured and used

PFE EMAC2

### 7.4.2 Prerequisites

Connect the P3B (EMAC2) port to your host PC and configure the PC's IPv4 address as 10.0.2.10 with a subnet mask of 255.255.255.0.

### 7.4.3 Functionality

This demonstrates the use of PFE in a Master-Slave scenario. HIF0 is assigned to the PFE master driver running on the Cortex-M7\_1 core, while HIF1 is used by the PFE slave driver running on the Cortex-A53\_2 core. Open two windows on the PC and ping the IP addresses of the PFE Master (10.0.2.2) and PFE Slave (10.0.2.21) respectively to check whether the ping task is running normally.



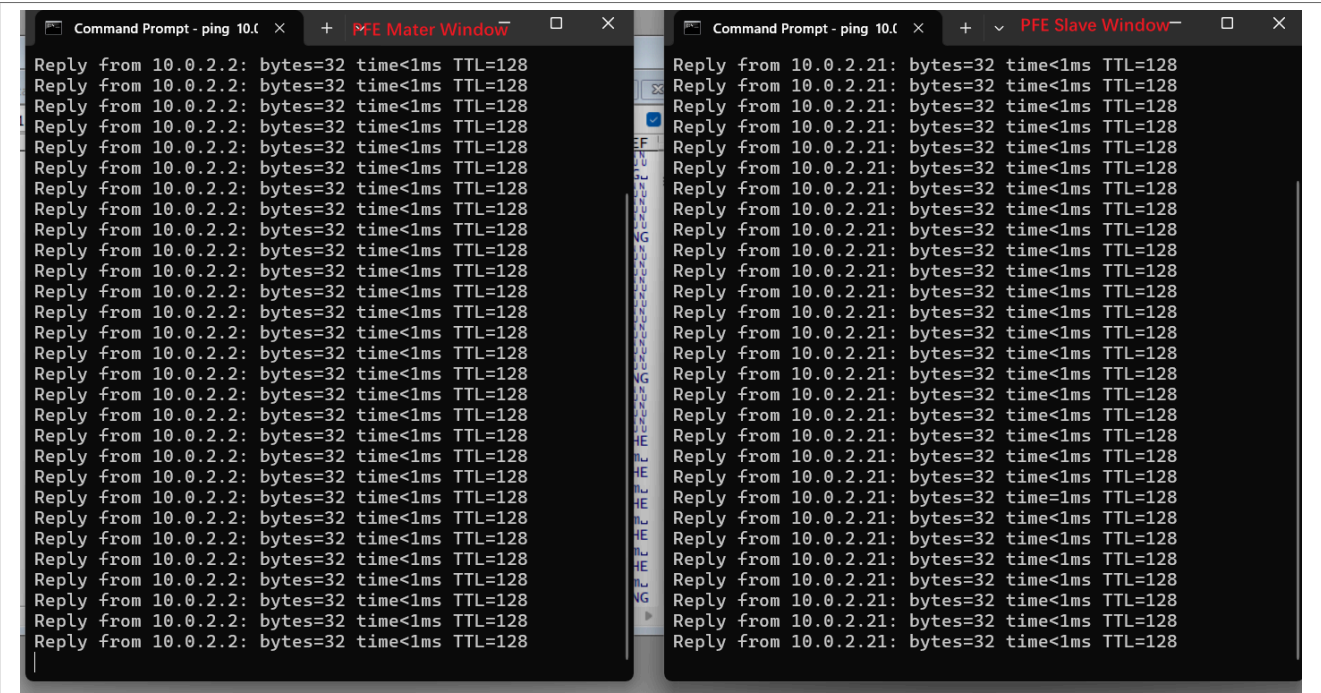


Figure 17. PFE Master and Slave ping task

## 8 S32G-VNP-RD2/RDB3 connectors

The second figure from the top shows the Ethernet interfaces.

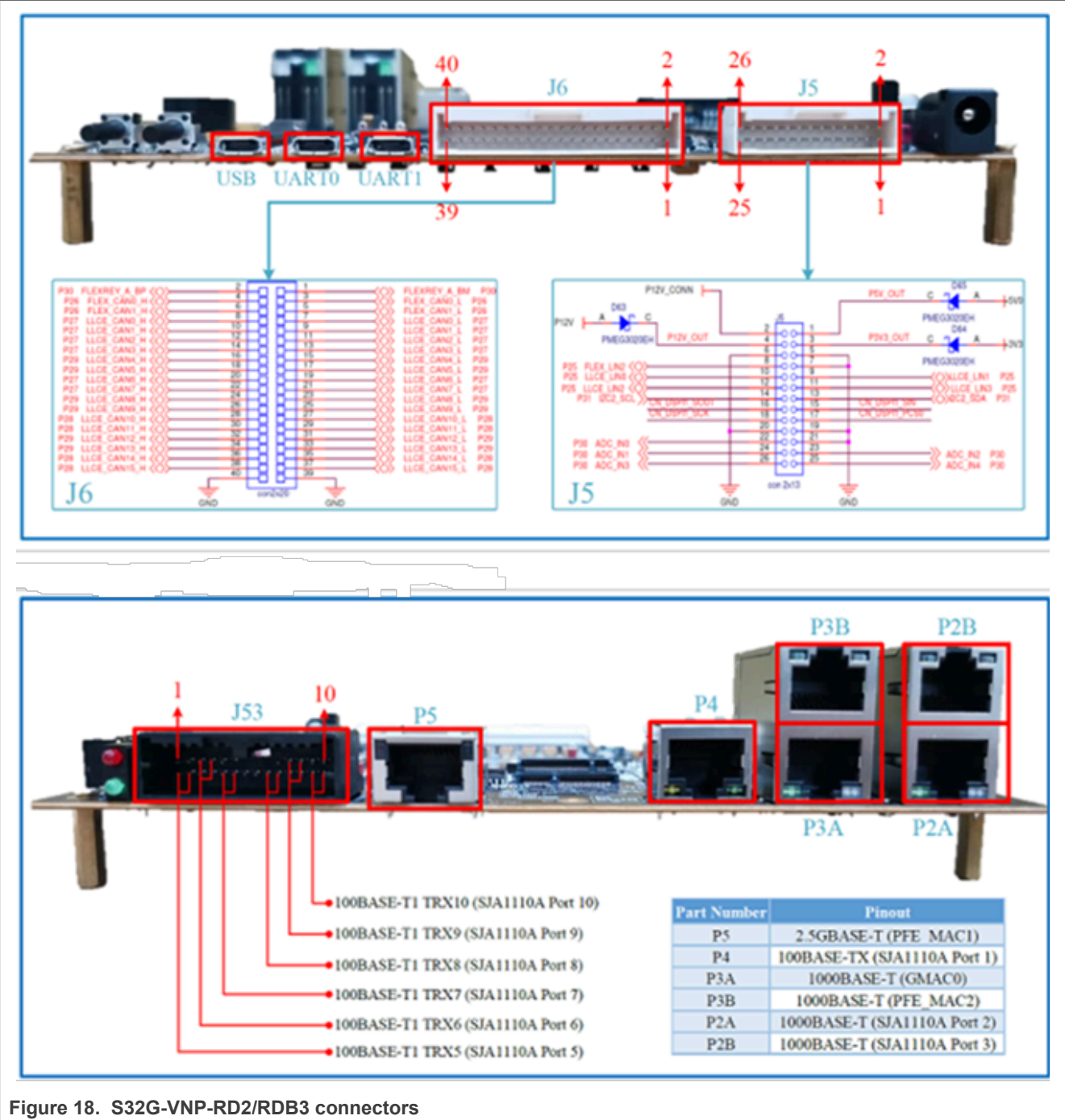


Figure 18. S32G-VNP-RD2/RDB3 connectors

## 9 Note about the source code in the document

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## 10 Revision history

Table 5. Revision history

| Document ID   | Release date | Description       |
|---------------|--------------|-------------------|
| UG10351 v.1.0 | 12 Nov 2025  | • Initial release |

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